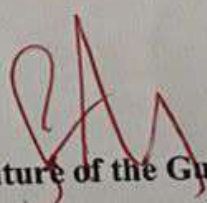
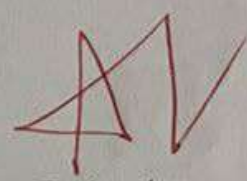


3. SOFTWARE REQUIREMENTS:

- Operating System : Windows / Linux / macOS
- Programming Language : Python 3.x
- IDE / Environment : Jupyter Notebook / Google Colab / VS Code
- Libraries : pandas, numpy, scikit-learn, scipy, matplotlib, seaborn
- Dataset Format : CSV (network flow / traffic data, e.g. CICIDS2017, NSL-KDD)
- Version Control : Git and GitHub



Signature of the Guide



Course Instructor

Name of Title: Malicious Traffic Pattern Clustering Using K-Means and Hierarchical Clustering

1. Abstract:

The rapid growth of network-connected devices and internet services has led to an enormous increase in network traffic volume, making manual identification of malicious activity increasingly difficult. Traditional signature-based intrusion detection systems rely on labeled attack data and struggle to detect previously unseen or evolving attack patterns. This project proposes an unsupervised approach for identifying malicious traffic patterns using K-Means and Hierarchical Clustering techniques, eliminating the dependency on labeled datasets while enabling the discovery of both known and novel traffic anomalies.

The proposed system uses a publicly available network traffic dataset, such as CICIDS2017 or NSL-KDD, containing flow-based features including packet size, duration, protocol type, source and destination ports, and byte counts. The data undergoes preprocessing steps including handling missing values, encoding categorical features, and feature scaling through standardization. K-Means Clustering is applied to partition traffic flows into distinct groups based on similarity, with the optimal number of clusters determined using the Elbow Method and Silhouette Score. Hierarchical (Agglomerative) Clustering is applied in parallel to construct a dendrogram, grouping traffic patterns without pre-specifying the number of clusters and revealing the nested structure of traffic behavior.

The resulting clusters are analyzed to distinguish benign traffic from malicious activity, with clusters exhibiting abnormal packet rates, unusual protocol usage, or atypical connection duration flagged as potentially malicious. The performance of the two clustering approaches is evaluated and compared using metrics such as the Silhouette Score, Davies-Bouldin Index, and cluster purity validated against known attack labels. A comparative analysis is presented to determine which clustering technique offers better separation and interpretability for malicious traffic detection. The expected outcome of this project is an unsupervised traffic clustering framework that can assist network administrators and security analysts in identifying suspicious traffic patterns and potential intrusions, while demonstrating the practical value of clustering techniques in cybersecurity applications.

2. INTRODUCTION:

Network security has become a critical concern as organizations increasingly rely on interconnected systems and internet-based services. Malicious activities such as denial-of-service attacks, port scanning, botnet communication, and data exfiltration generate distinct traffic patterns that differ from normal network behavior. Detecting such activity in real time is essential for preventing data breaches, service disruption, and financial loss. Traditional intrusion detection systems primarily depend on supervised learning models trained on labeled attack data, which require continuous updates as new attack types emerge and often fail to detect zero-day or previously unseen threats.

Clustering, an unsupervised machine learning technique, offers an alternative approach by grouping traffic flows based on inherent similarity in their features, without requiring labeled training data. This makes clustering particularly suitable for identifying novel or evolving attack patterns that supervised models may fail to recognize. K-Means Clustering partitions data into a predefined number of clusters by minimizing the variance within each cluster, making it computationally efficient for large-scale traffic datasets.

Hierarchical Clustering, on the other hand, builds a tree-like structure of nested clusters, providing greater flexibility in exploring relationships between traffic groups at different levels of granularity without requiring the number of clusters to be fixed in advance. This project explores and compares both clustering techniques for identifying malicious traffic patterns, aiming to evaluate their effectiveness in distinguishing benign traffic from potential security threats.

A PROJECT TITLE
ON
MALICIOUS TRAFFIC PATTERN CLUSTERING USING K-MEANS AND
HIERARCHICAL CLUSTERING TECHNIQUES

project report submitted in partial fulfilment of the requirements for the Course

MACHINE LEARNING
BACHELOR OF TECHNOLOGY

in
A.Y.: 2026-27

by
Name P. Sai Harshith Reddy roll 2520030375

Under the Esteemed guidance of

Shaik Asif
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CSE



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